

What is RadVax?

RadVax is an open-source vaccine that anyone can make. It is designed by a team of researchers connected to Harvard Medical School. There is a great deal of information on their website, radvac.org, written primarily for other scientists. This document covers basic information for the general public. For clarity, this document uses RaDVaC to refer to the group (Rapid Deployment Vaccine Collaborative), and RadVax to refer to the vaccine itself.

How does RadVax work?

RadVax is built on 3 well-established technologies: - **nasal delivery** - which provides immunity where the virus enters/leaves the body - **chitosan nanoparticles** - which stimulate the immune system to fight an invader - **lab-grown protein fragments** - which tell the immune system which invader to fight

Each of these technologies has been used safely for many years - RadVax is primarily a method of combining them. The method for combining the technologies uses simple techniques that are possible for anyone with basic laboratory skills.

Here is what each of the components do:

Nasal delivery

This just means spraying the virus directly into the nose (allergy sprays, and some flu vaccines use the same approach). Since no needles or injections are required, the **vaccine can be administered by anyone**. Nasal delivery also means that the system-wide immune response isn't as strong as with an injection. This makes the vaccine safer, but also means that it needs to be taken 3 or 4 times (instead of just 1 or 2 injections).

Chitosan nanoparticles

The chitosan nanoparticles serve two purposes. The first is as a transport system, which brings the protein fragments into the body. Because the nanoparticles are about the same size as the COVID virus, they travel to the same parts of the nasal tissues, bringing the protein fragments with them. The nanoparticles also help keep the protein fragments in place long enough for the

body to respond.

The second purpose of chitosan is as an "adjuvant"; an adjuvant is a substance that stimulates the immune system, telling the body that it needs to fight an invader. Chitosan has been used in clinical trials, and found safe enough that the EU has recommended it for use as an adjuvant in vaccines.

Lab-grown protein fragments

Your body learns to recognize invading viruses based on the shape of the virus' outer protein shell. So to teach the body what to fight, most COVID vaccines include small pieces of the virus' shell that are both distinct to the virus, and likely to be recognized by the immune system. These small pieces of protein are called "peptides". The world is full of peptides that we eat and breathe every day, without any problem since peptides themselves are generally harmless. They are made in the cells of plants and animals, and can't reproduce. It is only when a peptide is combined with an adjuvant that the body learns to attack.

One difference between the various COVID vaccines is how they introduce those peptides into the body. The Moderna, Pfizer, and AstraZeneca vaccines all trick the body into making these peptides. RadVax is a little bit different; it uses peptides that are made in a lab, and delivers them directly to the body (similar to the Novavax vaccine). While the delivery methods differ, the core principle is the same - **the combination of an adjuvant and the peptides teach the body what to attack.**

The choice of which peptides to use is one of the most important parts of any vaccine. Scientists make the decision based on several factors about peptides, including which peptides: - can be dissolved in water - so they can be delivered to the body - can be recognized by the immune system - so the body can learn what to attack - are unique to the virus - so the body doesn't become sensitive to anything besides the virus - are likely to remain unchanged as the virus mutates - so that the immunity is long-lasting

How safe is RadVax?

RadVax should be very, very safe.

Nasal delivery was chosen because it is a well-established, safe delivery method which has the additional benefit of focussing the immune response in the mucus membranes of the nose and sinuses. This means that the worst side effect is typically just a stuffy nose, or in a few cases a mild headache.

Chitosan nanoparticles as a delivery/adjuvant was chosen because it has been used in clinical trials, has an excellent safety record, and was already recommended for use by the EU. It stimulates the immune system, but doesn't stimulate it too much.

Lab-grown protein fragments (peptides) were selected carefully as well. Amongst other things, the scientists at RaDVaC have compared the peptide sequences against the entire human genome (using BLAST), to be sure that the peptides used won't be mistaken for some part of the body.

Possibly the best indication of its safety: all of the RaDVaC team has not only taken RadVax themselves, they've also offered it to their families.

How effective is RadVax?

It's not yet clear; the RaDVaC group is trying to find government partners to begin clinical trials for efficacy.

That said, there are strong theoretical reasons to believe that RadVax provides very good protection at the mucus membranes in the nasal area. This helps do two things: - prevent you from acquiring COVID (because the virus enters the body by breathing in) - prevent you from spreading COVID (because you breathe out virus growing in the nose)

Unfortunately, testing for this localized immune response is difficult. Some RadVaxers with the necessary equipment have done so, and reported signs of local immunity, which is both expected and encouraging.

The other difficulty in evaluating efficacy is that RadVax stimulates a different part of the immune system from most vaccines. The part of the immune system activated is determined primarily by the adjuvant; RadVax uses chitosan, which primarily activates T-cells. Most other vaccines instead primarily activate B-cells. B-cells work by identifying invading viruses, and then recruiting other parts of the immune system to attack. It is easy to test for B-cell activation, because they produce antibodies which circulate in the blood. T-cells, in contrast, don't directly attack the virus; instead they attack your own cells if they have been taken over by the virus. This makes T-cell response much harder to evaluate, even though it can be very effective.

How difficult is it to make RadVax?

For the latest generations, **it's very easy**. It takes only **basic laboratory skills** to make RadVax, so anyone with a bachelor's degree in chemistry, biology, or a related field can learn to make it in

about an hour. Once the ingredients are acquired, the process itself can be **done in less than an hour**.

There are **only 5 ingredients**: - distilled water - salt (sodium chloride) - TPP (trisodium phosphate, a widely-used food additive) - chitosan nanoparticles (ideally in the form HACCC, available from lab suppliers) - peptide sequences (hundreds of companies worldwide do custom sequencing)

Why hasn't RadVax been used before?

All of the individual components of RadVax **have** been used before. What's new is the way they're put together.

Nasal-delivery vaccines are widely used in lab and experimental settings. They are not commonly used for general vaccines, though, because they typically require multiple doses in order to build an immune response. One notable exception is the nasal flu vaccine. We can get away with a weak immune response in this case because basically everyone has had the flu before, so the body already has defenses against it... we just need to teach it about the new variants.

However, there are several challenges to using a nasal vaccine, including the RadVax, at scale. In rich/developed countries, time is at a premium, and it's difficult to get people to come to a doctor's office four times in order to complete a vaccination. In developing countries, on the other hand, time carries less cost, so multiple doses are less onerous.

Another challenge is the short shelf life of RadVax (around two weeks). Unlike the industrial vaccines, it can't be frozen, so it needs to be used fairly soon after being made. This is fine for use in a laboratory setting, but is not very effective at industrial scales (where it's better to have a shelf life measured in years). Partly because the shelf life is so short, RadVax is a labor-intensive option (as opposed to most commercial vaccines, which are capital-intensive). It needs to be made in relatively small batches, for use shortly after creation. So it will probably be made by individual clinics, for administration to their patients. This labor cost isn't affordable for rich countries, but is much more of an option in developing countries.

How does RadVax interact with the commercial vaccines?

You can and should take a commercial vaccine as soon as you are able to.

You can also take RadVax, as there should be no negative interactions between them. RadVax recruits a different part of the immune system than commercial vaccines (T-cells, as opposed to B-cells), so it is a complement, not a substitute. It will provide **additional protection** to you and your family that you can't get from commercial vaccines, so you should still **take both**.

In which ways is RadVax better than commercial vaccines?

RadVax has a few strengths that commercial vaccines don't, both on the personal level, but also on a public-policy level.

Personal benefits

Taking two different vaccines can produce **synergistic effects**, making RadVax + commercial a better choice than commercial alone: <https://www.npr.org/sections/health-shots/2021/05/05/993882203/giving-2-doses-of-different-covid-19-vaccine-could-boost-immune-response>

Unlike nasal-delivery vaccines, injections do not produce mucosal protection. So you can still acquire COVID, although the commercial (injection-based) vaccine will make it much easier for your body to fight the virus. Nasal-delivery vaccines can produce an immune response right at the point where the virus enters the body, which can **prevent you from catching COVID in the first place**.

Because RadVax is both safe and simple, it's also very flexible. The peptides used in RadVax have already been updated several times, both as scientists learn more about the virus, and as new variants evolve. RadVax is currently in its 10th generation, and is based on the latest knowledge about which peptides produce an immune response. It includes an optional peptide that should be **especially helpful for training the immune system of people who have Hispanic, Southeast Asian, or Native American ancestry**.

Most commercial vaccines for COVID are based on very early research. They use peptides that are part of the spike portion of the shell (which is what the virus uses to enter our cells). The advantage at the time was that scientists could be sure it was distinct to the virus, even early on when we didn't know much at all about how COVID attacks us. The disadvantage is that this is one of the things most likely to change as the virus evolves, leading to "vaccine escape".

Unfortunately, our medical regulatory system doesn't easily allow for changes in the formulas used. Right now, the **COVID virus is evolving faster than our regulatory system can keep**

up with it.

RadVax was based on later research, and uses a slightly different approach. First, rather than focussing on the spike protein, the RaDVaC team compared SARS-COV-2 (the COVID virus) to SARS-COV-1 (the original SARS from 2002), to see **which parts of the virus had remained unchanged** over 2 decades. They also looked at the immune response of people who had acquired SARS-COV-1, to see which parts of the virus they were still immune to. They used this information to choose peptides that should both **prevent vaccine escape**, and provide **long-lived immunity**.

And while RadVax doesn't focus on the spike protein, RadVax is certainly able to use peptides related to the spike protein. The latest generation (Gen 10) of RadVax does have an optional peptide that should help **defend against the new, more dangerous variants** (Brazilian strain P.1, English strain B.1.1.7, and South African strain 501Y.V2). They hope to soon have a peptide to address the Indian strain B.1.167.

Public-policy benefits

RadVax is especially useful for developing countries, because it uses a different method for delivery, and thus has an **entirely different supply chain**. The supply chain for commercial vaccines has already been stretched so thin that production of standard childhood vaccines such as MMR, or polio has nearly stopped. In contrast, the supply chain for RadVax can be used without creating further health risks.

It also provides **vaccine equity**. Currently the industrialized nations are first in line for the commercial vaccines, so commercial vaccines won't be available for poor nations anytime soon. For example, the USA is holding back over 30 million doses of AstraZeneca (enough to vaccinate the entire country of Peru). In addition, commercial producers sometimes renege on their commitments to poor countries (Pfizer backed out of an agreement to supply 10 million doses to Colombia). RadVax can be produced **locally in-country**, since it only relies on a few ingredients, which are widely available. **Countries can begin making RadVax themselves**, without relying on international corporations which may or may not come through on their commitments.

Can RadVax be used for special populations?

Eldery persons

There should be no concerns here. Many of the people who have taken RadVax already are well into their 80s, and haven't reported any unusual side effects.

Pregnant women & breastfeeding mothers

We should always be especially cautious when considering nursing or pregnant mothers. Primarily because at that age, children are developing so rapidly that small effects can be harmful in later years. But in this case, there's no reason to believe there's any particular risk. The immune response is mild (no fevers reported, etc), and concentrated in the nasal tissues. It's certainly less dangerous than acquiring COVID.

If you have the luxury of living in a country which is already reaching herd immunity, then the safer option might be to forgo RadVax. But if you're in a country where you still have a high chance of getting COVID, then check with your doctor to see if RadVax might be the better choice for you.

Children

There's no reason to believe there's any particular risk with administering RadVax to children. But it's extremely difficult to get a small child to breathe in deeply while nasal spray enters their nose, and therefore challenging to administer effectively.